

power

File: sheets/power.kicad_sch

thermocouples

File: sheets/thermocouples.kicad_sch

io

File: sheets/io.kicad_sch

test

File: sheets/test.kicad_sch

mcu

File: sheets/mcu.kicad_sch

ssr

File: sheets/ssr.kicad_sch

ct

File: sheets/ct.kicad_sch

NOTES — Bisque kiln controller, rev B

SSR terminals J4/J9: pin1 = VIN_P (24 V, fused and TVS-clamped, ALWAYS LIVE), pin2 = the switched low side. The watchdog gates the shared return (Q7), not the supply, so a tripped watchdog leaves 24 V on pin 1 and opens pin 2. Boot-safe: R7/R20 hold each gate down while the optos are dark. The optos are NOT an isolation barrier – both sides share GND. Rev B's opto-ISOLATED design was reverted and stays reverted; U8/U9 are here to keep a 24 V gate fault off the GPIO. Per channel <=200 mA: F1 holds at 750 mA and U11 takes ~250.

TC1 terminal J3 / TC2 terminal J8: pin1 = K+, pin2 = K– – both float, biased near AGND only through each MAX31856's internal BIAS network. Ungrounded-junction probes required: two grounded-junction probes in one kiln would loop through it.

Nav switch J6 is panel-mounted; inputs use ESP32 pull-ups.

J11: IN1/IN2/IN3 (lid/gas-flow/spare) + GND – dry contact each channel to GND.

Display J5 (14-pin, ST7796S + XPT2046 touch module, LCDWIKI MSP4021): 1=+5V 2=GND 3=CS 4=RST 5=DC 6=SDI/MOSI 7=SCK 8=BL 9=SDO/MISO 10=T_CLK 11=T_CS 12=T_DIN 13=T_DO 14=T_IRQ. PIN 1 IS +5V, NOT 3V3 – the module regulates on board; do not wire a 3.3V-only panel to it. Logic is 3.3V; touch shares SPI2 with the LCD and both MAX31856s (R39–R43 damping).

J12: CT current inputs, CTA_P/CTA_N/CTB_P/CTB_N – one CT clamp per SSR zone.

J13: DNP SELV voltage sense header. NOT mains-rated, not fitted – do not wire to AC mains. Y1/C25/C26 load caps are an ASSUMED value (unverified C_L), see design.py comment. ADE7953 I2C address 0x38 collides with PCF8574A.

4-layer, 100 x 100 mm, JLCPCB standard process
ESP32-S3-WROOM-1U-N16R2 + 2x MAX31856 + dual SSR + ADE7953
Bisque project

Sheet: /
File: bisque-controller.kicad_sch

Title: Bisque Kiln Controller

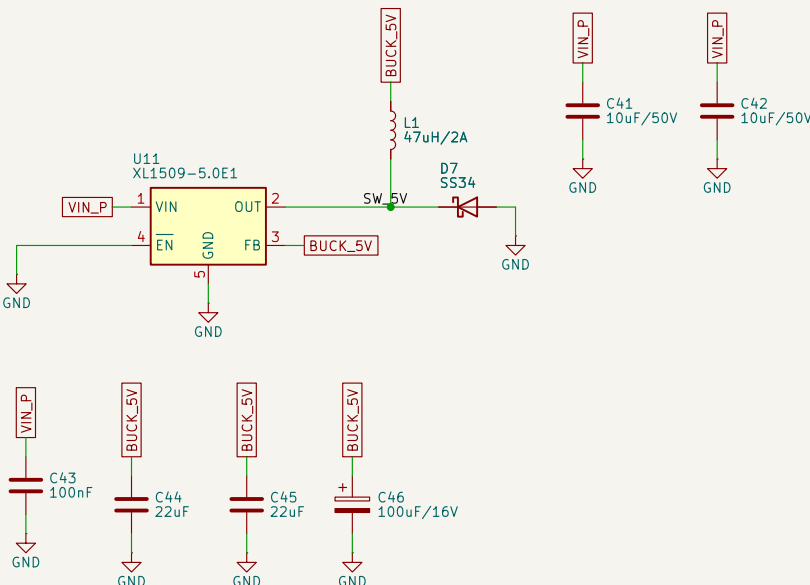
Size: A3Date: 2026-09-21

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Rev: B.2Id: 1/8

F

24 V -> 5 V BUCK XL1509-5.0, fixed output
C41-C43 in, D7 catch, L1 47uH, C44/C45/C46 out. FB ties to the
output because the part is the fixed 5.0 V version; EN is
active LOW and grounded. C46 is an ELECTROLYTIC on purpose:
voltage-mode with fixed compensation needs its ESR zero, and
an all-ceramic output has none. Keep C41/C42 - U11 - D7
tight; that loop, not L1, is what radiates.



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A



S



Title: Bisque Kiln Controller – ESP32-S3, USB, reset,

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A



A



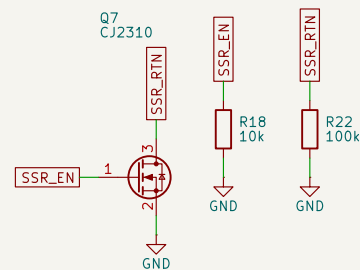
C

D

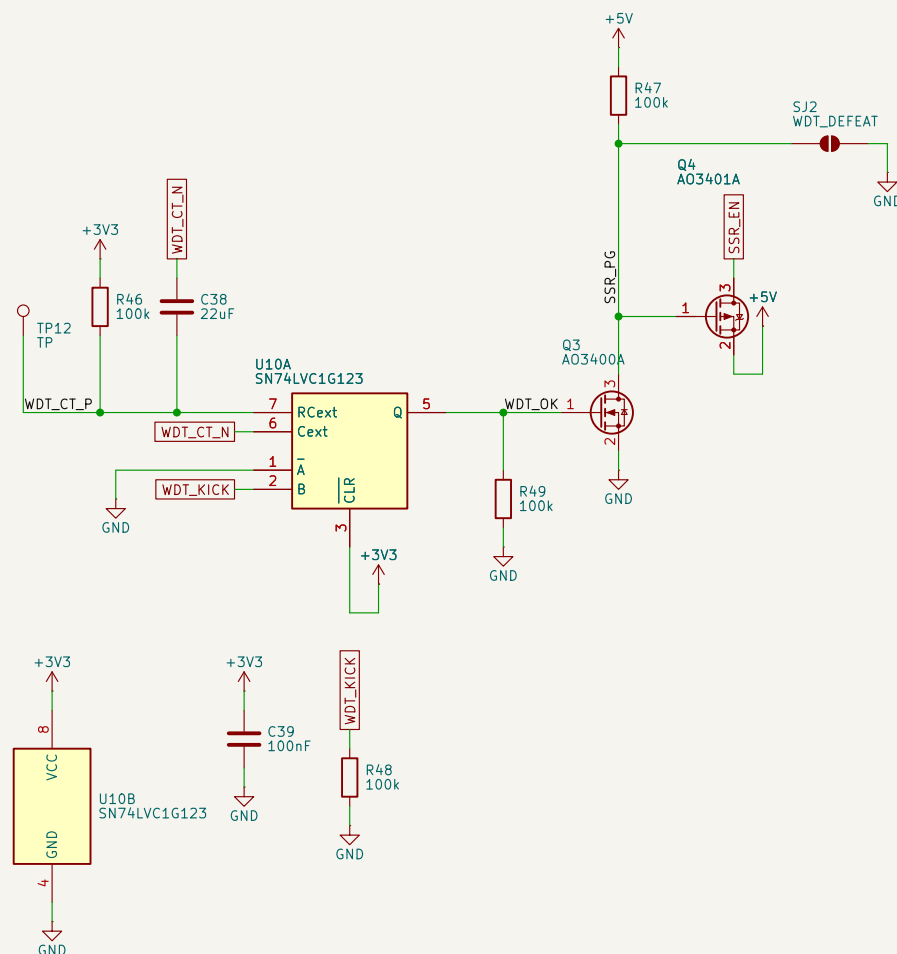
E

11

24 V terminals off VIN_P, switched low side by a 60 V CJ2310. The GPIO drives an opto LED, never the gate: at 24 V a gate-to-drain failure would otherwise put 240 mA into the pin. Opto collectors sit on SSR_EN (5 V), so the gate lands at ~4.9 V from the RAIL rather than from CTR. Q7 is the watchdog's gate in the shared return – it opens a channel whose MOSFET failed short. R18/R22 define SSR_EN and SSR_RTN when their switches are off. NOT an isolation barrier – see design.py's SSR block.

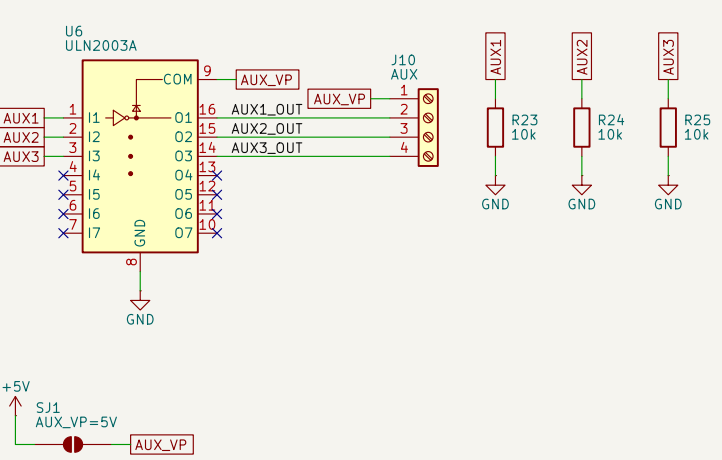


U10 retriggerable one-shot: every rising edge on WDT_KICK restarts a ~2.2 s window, so Q stays high only while firmware keeps kicking. Q3 holds Q3 on; Q3 pulls SSR_PG down, turning on high-side Q4, which supplies SSR_EN – the +5V GATE DRIVE rail feeding both opto collectors and Q7's gate. R46/C38 set the window. R48 holds B low when the MCU is high-Z. R47 is the fail-safe pull-up. S12 = bring-up defeat, REMOVE for service.

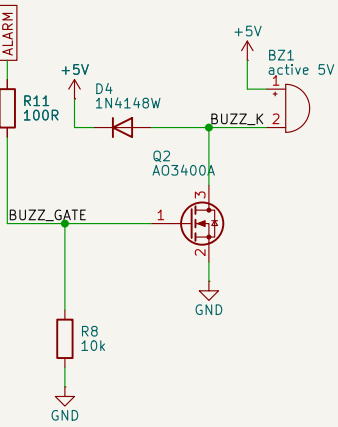


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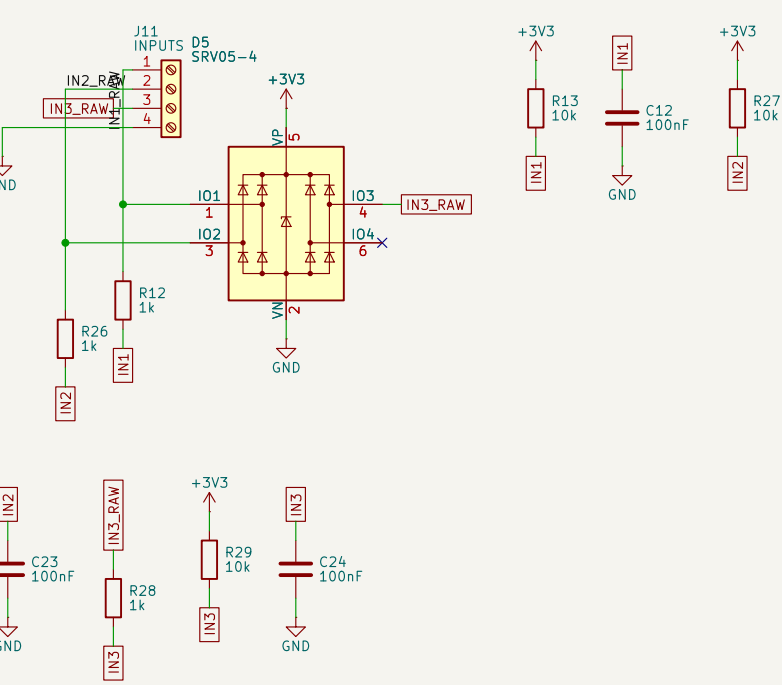
AUX OUTPUT BANK
ULN2003 vent/purge/spare; COM->AUX_VP, SJ1 links to +5V



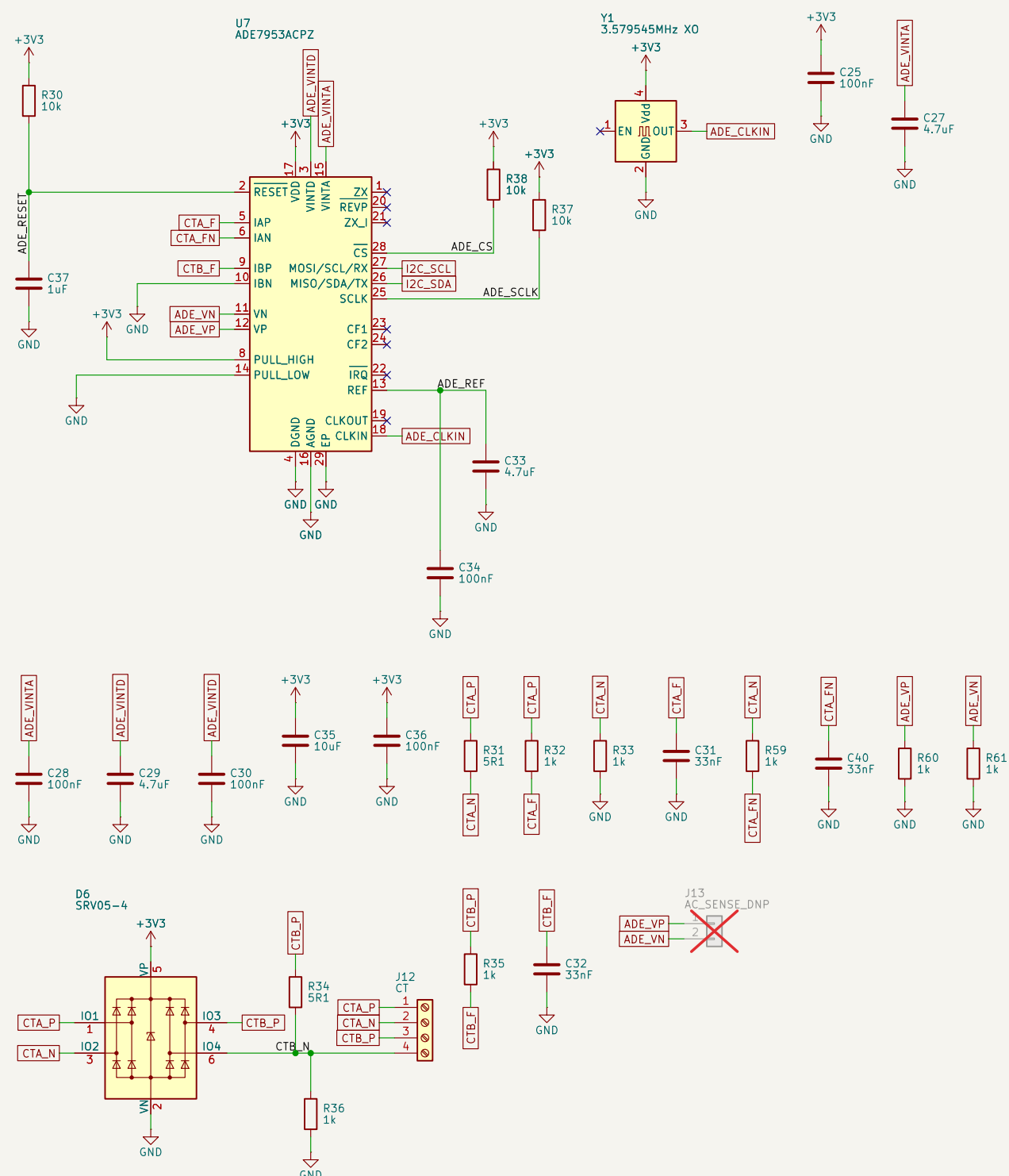
ALARM BUZZER



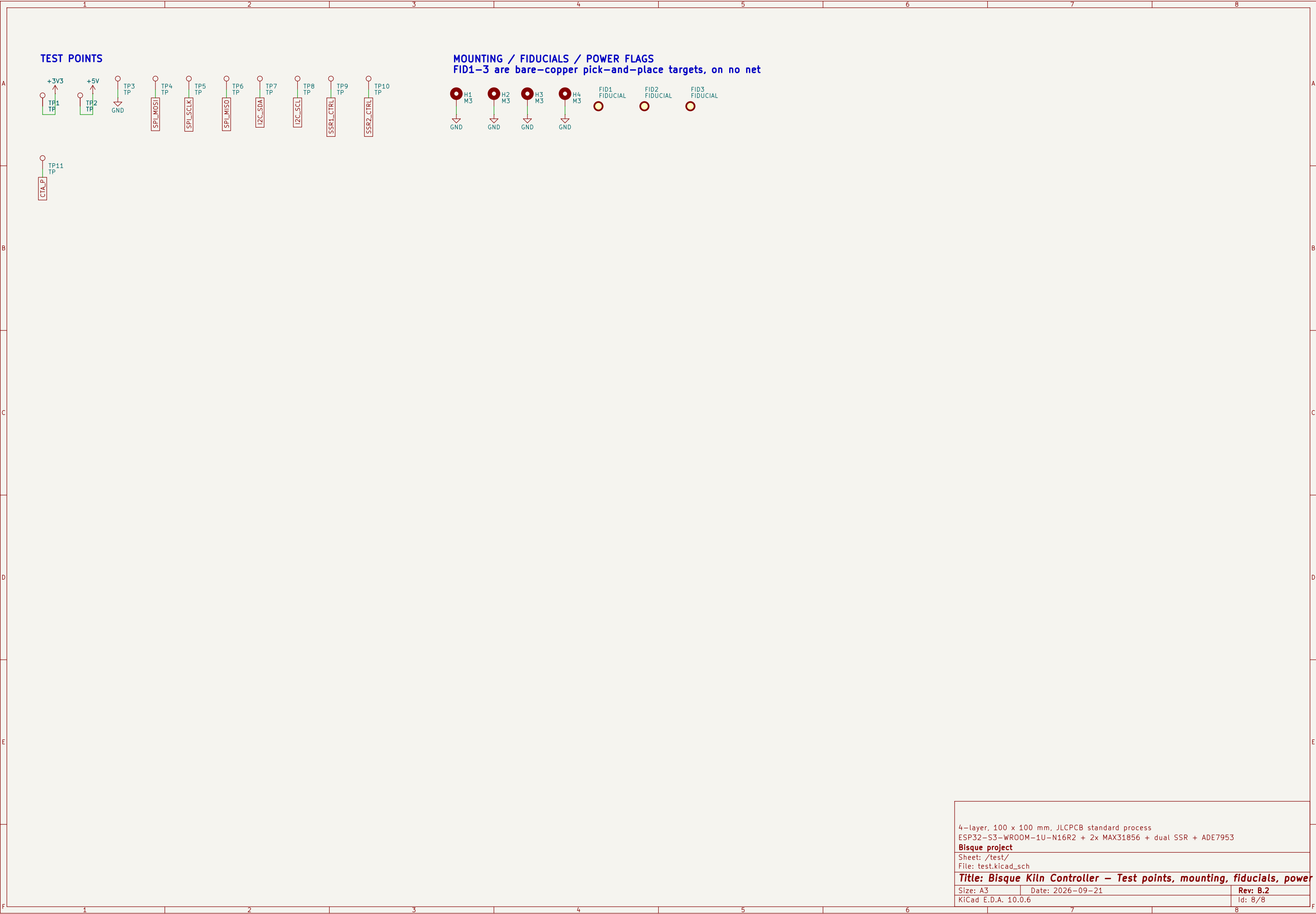
PROTECTED INPUTS x3 lid / gas-flow / spare
1k + 10k pull-up + 100nF each, SRV05-4 TVS



CT CURRENT SENSING ADE7953, I2C, current-only
no mains – VP/VN to DNP J13



4-layer, 100 x 100 mm, JLCPCB standard process
 E5P32-S3-WR00M-1U-N16R2 + 2x MAX31856 + dual SSR + ADE7953
Bisque project
 Sheet: /ct/
 File: ct.kicad_sch
Title: Bisque Kiln Controller - CT current sensing (ADE7953)
 Size: A2 Date: 2026-09-21 Rev: 6.2
 KiCad E.D.A. 10.0.6 Rev: 7/8



4-layer, 100 x 100 mm, JLCPCB standard process

ESP32-S3-WROOM-1U-N16R2 + 2x MAX31856 + dual SSR + ADE7953

Bisque project

Sheet: /test/

File: test.kicad_sch

Title: Bisque Kiln Controller – Test points, mounting, fiducials, power flags

Size: A3

Date: 2026-09-21

Rev: B.2

KiCad E.D.A. 10.0.6

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